

Digital Forensics Case Report: *Operation Hollow Invoice*

CASE ID: 005

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CLIENT CONTEXT: Small Accounting Firm

1. Executive Summary

An investigation was initiated following suspicious activity at an accounting firm, culminating in an unauthorized 2:00 AM access to the firm's financial system. The incident originated from an employee executing an email attachment masquerading as a standard quarterly invoice.

Static file analysis using hex editing techniques revealed a multi-stage intrusion. The initial attachment was a disguised executable that likely deployed a keylogger. This payload subsequently captured financial system credentials and staged sensitive internal documents for exfiltration by concealing them within a deceptive image file.

2. Case Background & Scope

The firm's IT Administrator observed unauthorized network access three days after an employee interacted with a suspicious email attachment ([invoice_Q3_2024.pdf](#)). During preliminary triage, the IT Admin secured the original attachment and two anomalous files discovered on the employee's workstation.

Objective: Conduct static analysis on the three extracted files using only direct hex examination (HxD) to determine their true file types, uncover hidden contents, and construct a narrative of the attacker's actions.

Evidence Processed:

1. [invoice_Q3_2024.pdf](#) (Initial email attachment)
2. [system_update.exe](#) (Found on workstation)
3. [photo_family.jpg](#) (Found on workstation)

3. Methodology & Analysis

The examination relied exclusively on manual hex editing (HxD). No automated file scanners or forensic suites (e.g., Autopsy) were utilized. The analysis focused on verifying magic bytes (file signatures), identifying ASCII

string anomalies, calculating offsets, and exposing anti-forensic techniques such as file extension masquerading and evidence hiding.

4. Key Findings

Exhibit A: The Initial Vector

- **File Name:** invoice_Q3_2024.pdf
- **Claimed Type:** Adobe PDF (.pdf)
- **True Type:** Windows Executable (.exe / .dll)

16																Windows (ANSI)	hex
Recommendation_Maxwell.pdf																invoice_Q3_2024.pdf	
Offset(h)	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	Decoded text
00000000	4D	5A	50	20	54	68	69	73	20	66	69	6C	65	20	69	73	MZ This file is
00000010	20	6E	6F	74	20	77	68	61	74	20	69	74	20	63	6C	61	not what it cla
00000020	69	6D	73	20	74	6F	20	62	65	2E	0D	0A	54	68	65	20	ims to be...The
00000030	65	6D	70	6C	6F	79	65	65	20	6F	70	65	6E	65	64	20	employee opened
00000040	74	68	69	73	20	61	74	20	30	39	3A	31	34	20	6F	6E	this at 09:14 on
00000050	20	61	20	54	75	65	73	64	61	79	2E	0D	0A	43	6F	6E	a Tuesday...Con
00000060	66	69	64	65	6E	74	69	61	6C	3A	20	50	72	6F	6A	65	fidential: Proje
00000070	63	74	20	4E	49	47	48	54	46	41	4C	4C	20	74	72	61	ct NIGHTFALL tra

Technical Analysis: Analysis of the file header revealed the hex bytes 4D 5A (ASCII: MZ), the standard signature for a Windows executable, completely contradicting the .pdf extension. Furthermore, unlike a genuine PDF which contains compressed binary data, the file’s payload was readable in plain text.

A string search revealed critical evidence at offset 0x73 (decimal 115):

NIGHTFALL transfer approved

00000060	66	69	64	65	6E	74	69	61	6C	3A	20	50	72	6F	6A	65	fidential: Proje
00000070	63	74	20	4E	49	47	48	54	46	41	4C	4C	20	74	72	61	ct NIGHTFALL tra
00000080	6E	73	68	65	72	20	61	70	70	72	6F	6F	65	64	2E	0D	nsfer approved..
00000090	0A	41	6D	6F	75	6E	74	3A	20	24	38	34	37	2C	30	30	.Amount: \$847,00
000000A0	30	2E	40	52	65	63	69	70	69	65	6E	74	3A	20	41	63	0. Recipient: Ac
000000B0	63	68	75	6E	74	20	32	39	34	3E	2D	58	58	58	58	2E	count 2940-XXXX.
000000C0	0D	0A	41	75	74	69	6F	72	69	6A	61	74	69	6F	6E	20	..Authorization
000000D0	63	6F	64	65	3A	20	4E	46	20	32	30	32	34	2D	43	4F	code: NF-2024-CO
000000E0	4E	46	49	52	4D	2E											NFIRM.

Checksum	Search (Units)
Offset	Excerpt (hex)
73	65 6E 74 69 61 6C 3A 20 50 72 6F 6A 65 63 74 20 4E 49 47 48 54 46 41 4C 4C 20 74 72 61 6E 73 66
	Excerpt (text)
	ential: Project NIGHTFALL transf

Significance: The attacker successfully bypassed initial suspicion by masquerading a malicious executable as a standard business document.

Exhibit B: Credential Harvesting

- **File Name:** `system_update.exe`
- **Claimed Type:** Windows Executable (`.exe`)
- **True Type:** Windows Registry Hive (`.regf`)

Technical Analysis: The first four bytes of the file were `52 45 47 46` (ASCII: `REGF`), indicating this was not an application update, but a Windows Registry file. String analysis exposed keylogging activity targeting the local Windows user `JSMITH` (found at offset `0x25`).

Further inspection at offset `0xAF` uncovered captured plain-text credentials:

Target: `banking.firstnational.com` **Username:** `jsmith_finance` **Password:** `Acc0unting2024`

Timestamp Discrepancy: The file contained a hardcoded timestamp of `2024-09-17 02:14:33`. Given the email was reportedly opened at 09:14, the 7-hour discrepancy indicates either a timezone offset (e.g., the system recording in UTC while the local time was UTC-7) or deliberate timestamp obfuscation by the malware.

Offset(h)	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	Decoded text
00000000	52	45	47	46	20	54	68	69	73	20	69	73	20	61	20	72	REGF This is a r
00000010	65	67	69	73	74	72	79	20	66	69	6C	65	2E	0D	0A	55	egistry file...U
00000020	73	65	72	3A	20	4A	53	4D	49	54	48	0D	0A	4C	61	73	ser: JSMITH..Las
00000030	74	20	61	63	74	69	76	65	3A	20	32	30	32	34	2D	30	t active: 2024-0
00000040	39	2D	31	37	20	30	32	3A	31	34	3A	33	33	0D	0A	4B	9-17 02:14:33..K
00000050	65	79	6C	6F	67	65	72	20	73	65	73	73	69	6F	6E		eylogger session
00000060	20	64	61	74	61	20	66	6F	6C	6C	6F	77	73	2E	0D	0A	data follows...
00000070	43	61	70	74	7E	73	65	64	3A	20	69	61	6F	6B	6B	6F	Captured: bankin

Significance: This file confirms the attacker's motive: financial credential theft. The malware successfully logged keystrokes and stored them in a disguised registry hive to evade basic antivirus detection.

Exhibit C: Data Staging & Concealment

- **File Name:** `photo_family.jpg`
- **Claimed Type:** JPEG Image (`.jpg`)
- **True Type:** ZIP Archive appended to a Text header

Technical Analysis: The opening bytes lacked any recognized image file signature. Instead, the first 20 bytes resolved to plain text: `This is normal image content. Nothing to see here.` A genuine JPEG contains compressed binary data, not plain English.

At offset **0x33**, the hex bytes **50 4B** (ASCII: **PK**) were discovered. This is the magic byte signature for a ZIP archive (named after Phil Katz). Extracting the data following the **PK** header revealed a hidden archive containing three highly sensitive documents:

1. **NF_transfer_authorization.docx**
2. **account_routing_offshore.xlsx**
3. **personal_id_scan.pdf**

Significance: The attacker utilized a classic anti-forensic technique—evidence hiding via appended data. By packaging stolen documents inside a file with a **.jpg** extension, standard image viewers would simply return a read error, while automated network scanners might ignore the file as benign media during exfiltration.

5. Conclusion & Limitations

Incident Narrative: Operation Hollow Invoice was a targeted phishing attack designed for financial theft and data exfiltration. The intrusion began with a malicious executable disguised as a PDF. Upon execution, the payload deployed a keylogger (disguised as an **.exe** file but functioning as a registry hive) that successfully captured the firm's banking credentials. Finally, the attacker aggregated sensitive offshore routing and authorization documents, zipping them inside a fake JPEG to prepare for stealthy exfiltration.

Limitations of Analysis: This preliminary report is limited strictly to static hex analysis. While it effectively proves intent, file structures, and data compromise, it lacks dynamic execution context.

Recommended Next Steps: To fully resolve this incident, I recommend:

1. **MFT & Timeline Analysis:** Parse the Master File Table (MFT) to resolve the 7-hour timestamp discrepancy and determine the exact creation/modification times of these artifacts.
2. **Network Forensics:** Review firewall and proxy logs for outbound connections to **banking.firstnational.com** and unauthorized data transfers corresponding to the zipped documents.
3. **Endpoint Remediation:** Immediately isolate **JSMITH**'s workstation and initiate a password reset protocol for all associated financial accounts.